

How Consistency Factors in Retail Shift Scheduling Affect Productivity

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1. Introduction

About 26 million Americans work shift jobs, and retail accounts for a large share of them. These workers do not have fixed schedules the way salaried employees do. Instead, their hours get set week to week, often just days in advance, based on whatever consumer demand looks like at that moment. Researchers call this just-in-time scheduling, and it is common practice across the retail sector (Lambert, 2008). Nearly 90% of hourly retail workers deal with some form of schedule instability because of this (Lambert et al., 2014). That instability has real costs for workers. Schneider and Harknett (2019) surveyed workers across a range of large retail employers and found that unpredictable scheduling does more damage to worker well-being than low wages. The main culprits were sleep disruption and the difficulty of managing family and personal obligations when you do not know your schedule more than a few days out. A follow-up study by Harknett et al. (2021) looked at what happened in Seattle after the city passed a Secure Scheduling Ordinance in 2017, which required retail employers to post schedules at least two weeks ahead. Workers in Seattle showed better sleep, less financial stress, and higher overall well-being compared to workers at the same companies in cities without the law. The human cost of unstable scheduling is well established at this point.

On the employer side, there are many objections to policy regarding shift scheduling predictability and consistency. Retail executives have long argued that restricting how and when schedules can be set limits their ability to respond to shifting consumer demand, and that requiring more stable schedules would hurt the operational efficiency of these businesses. However, the evidence does not clearly support that concern. For example, a field experiment at Gap Inc. tested a bundle of scheduling improvements that included greater consistency and found that store labor productivity rose by 5.1% and sales rose by 3.3% (Kesavan et al., 2022). That study, however, tested a bundle that mixed consistency, advance notice, worker input, and other changes together, so it is not possible to know which piece drove that gain in labor productivity. Nonetheless, policy has moved ahead of the research, as supported by the Fair Workweek legislation that now applies to more than 1.8 million workers in places like San Francisco,

New York City, Oregon, and Seattle (Zipperer and Shierholz, 2018). These laws require advance notice and restrict last-minute changes to shift workers' schedules, but they are written quite broadly. They treat schedule stability as one variable rather than asking which dimension of stability matters most, and they say nothing about how employers should allocate their limited scheduling flexibility when they cannot stabilize everything at once.

Lu et al. (2022) made the first attempt to address that gap at the individual productivity level, isolating schedule consistency and measuring its effect on worker output at the transaction level. They identified two distinct dimensions: hour-of-the-day consistency (HOTD), capturing whether a worker tends to be scheduled in the same hourly slot across days, and day-of-the-week consistency (DOTW), capturing whether a worker tends to be scheduled on the same days across weeks. Both independently raised cashier productivity. Shang et al. (2026) extended this framework to a warehouse setting with night shifts and found that DOTW produces gains four to eight times larger than HOTD, that HOTD's benefit disappears during sleep hours, and that DOTW is more stable across environmental conditions. Together these studies established that the two dimensions are not equivalent, but neither made the relationship between them a primary focus.

That is the gap this paper addresses. Lu et al. (2022) reported a negative and significant HOTD x DOTW interaction as a secondary finding, suggesting the two dimensions substitute for each other rather than compound. However, that result appeared in one paragraph with no formal hypothesis and no managerial development. Shang et al. (2026) did not test the interaction at all. If the two dimensions truly substitute, and if DOTW is already the stronger lever, managers should prioritize DOTW first, but no evidence-based guidance currently supports that decision. Fair Workweek laws do not make the distinction, and without understanding the interaction structure, managers are making these allocation choices without empirical grounding.

This paper makes the HOTD x DOTW interaction its central research question, using transaction-level data from a daytime grocery retail setting where the sleep-hour complication identified by Shang et al. (2026) does not apply. We estimate fixed-effects regression models with a multiplicative interaction term and interpret the interaction coefficient to assess direction and magnitude across varying levels of each consistency dimension. This paper has three goals. First, we provide the first confirmatory test of the interaction as a pre-specified hypothesis. Second, we offer a theoretical account of why the interaction might be negative and potentially asymmetric. DOTW operates through what we term community rhythm, the shared social structure of the week that most workers experience similarly regardless of individual differences. HOTD operates through what we term the endogenous circadian clock, the body's internal biological timer, which varies more across individuals based on chronotype, meaning a person's genetically influenced tendency toward earlier or later sleep-wake timing (Gentry et al., 2021). Because DOTW anchors a more universal rhythm, it tends to be the stronger lever, and once it is established, the marginal gain from adding HOTD consistency shrinks. Third, we translate these findings into practical scheduling guidance for managers, identifying which dimension to prioritize first and how the substitute relationship shapes the return from investing in each.

2. Literature Review

We will review several research papers on worker consequences of scheduling instability, schedule consistency and productivity, and the mechanisms on why consistency improves performance.

2.1 Worker Consequences of Schedule Instability

Studying how schedule instability affects workers is difficult without detailed data on individual scheduling practices. Survey-based approaches have helped fill this gap, and the Shift Project is one of the more comprehensive examples. It is a national survey that collects data through web-based surveys (Schneider & Harknett, 2019). The study used targeted advertisements on Facebook to target specific employers. Their methodology is slightly different on how they measure work schedule consistency, but

their results on psychological distress (Kessler K6), good sleep (Pittsburg Sleep Quality Index), and happiness highlight how schedule inconsistencies affect workers.

Schneider and Harknett (2019) found consistently through their measures that unstable and unpredictable scheduling is positively associated with psychological distress. Sleep quality was found to be negatively associated with their week-to-week variability in work hours. Additionally, they found there was a strong relationship between happiness and exposure to canceled shifts, on-call work, clopening, and exposure to unstable and unpredictable scheduling practices. The study, however, has some limitations of its own. Their findings establish that schedule inconsistency has measurable negative effects on workers. Their survey-based self-reports measure predictability and advance notice rather than our focus of hour-of-the-day or day-of-the-week consistency measures.

In a subsequent study, the implementation of fair workweek legislation in Seattle and its effect on workers is analyzed (Harknett et al., 2021). Harknett et al. (2021) examined Seattle's fair workweek law, which was enacted in 2017 to require two weeks' notice of schedules. Using a two-year comparison between Seattle workers and same-employer workers in cities without such laws, the study found that greater schedule predictability improves worker well-being and may enhance efficiency, providing evidence that scheduling legislation yields measurable benefits.

Predictability laws are proven to work and help workers, yet their implications on different consistency measures such as ours is still ambiguous. Sources such as the Economic Policy Institute, "Fair workweek laws help more than 1.8 million workers", documents the spread of fair workweek legislation in several locations such as San Francisco, New York City, Oregon, and Seattle. In this paper, fair scheduling is defined as one to two weeks notice. Current legislation is targeting schedule predictability broadly but does not make the distinction we make in this paper about consistency between HOTD, DOTW, and interaction effect. Our paper will inform what these laws should prioritize.

2.2 Schedule Consistency and Productivity

Several relevant studies help establish the relationship between schedule consistency and worker productivity, and provide frameworks for analyzing it. Lambert et al. (2019) examined how schedule predictability implementation affected store operations. For the purpose of our paper, we will focus on their schedule consistency and productivity framework. For their experiment conditions, the managers were instructed to post weekly schedules for a full month at a time. Managers in the control condition were told to continue with their usual practice of scheduling less than a week before the upcoming workweek.

Key findings relevant to our paper show that posting further in advance did not meaningfully improve workers' ability to anticipate hours since managers changed posted schedules frequently (Lambert et al., 2019). This emphasizes the difference between predictability and consistency, a key distinction we will directly build upon. In this paper we will focus more on the consistency portion of scheduling.

We previously addressed how businesses implement just-in-time scheduling practices and how this might adversely affect workers. We are interested in worker productivity and in order to measure it we need to operationalize just-in-time schedules in order to analyze them. Similar studies have tried to calculate how just-in-time scheduling affected productivity (Kamalahmadi et al., 2021). In this 25-store casual dining restaurant chain paper, they study how just-in-time scheduling impacts worker productivity. They consider two types of just-in-time schedules, one of them is scheduling before the day of service, and the other is scheduling the day of service. They operationalized schedule predictability via three dummies for real-time, short notice, and no-change schedules.

Interestingly, it was discovered that from their findings that short notice schedules did not harm productivity overall. However, real-time schedules affected overall worker productivity by 4.4%. Even though this paper helps us understand how to operationalize our paper, it tests predictability, rather than schedule consistency. This study helps complement ours as we will build upon the consistency aspect of scheduling.

We will now focus on research that has demonstrated that schedule consistency has improved financial performance. From their four dimension tests, they discovered that these bundled scheduling practices increase store productivity by 5.1%, while increasing sales and reducing labor (Kesavan et al., 2022). However, this is very limiting as the bundled practices impede us from isolating the effect of consistency alone. Additionally, they do not distinguish between hour-of-the-day from day-of-the-week contributions. This is where our study comes in handy as we are addressing this at the individual transaction level.

The study this paper builds upon used transaction-level data from two grocery store locations (Lu et al., 2022). Their data includes 1,238,418 transactions by 126 cashiers from 2016 to 2017. Their consistency variables are calculated in a 28-day rolling window. Schedule consistency for hour-of-the-day means how many days in the last 28 days has that employee worked at that particular hour. For day-of-the-week it means how many days in the last 28 days has this employee worked that same day of week. We will replicate their operationalization as we will validate their results and from there build further the interaction effect.

Lu et al. (2022) found that both the hour-of-the-day variable and the day-of-the-week variable are positive and significant. When a cashier works in an hour of the day slot that they have worked over the past four calendar weeks, they are on average 0.95% more productive than when they work in an hour of day slot during which they have not worked in the past four calendar weeks. Similarly, when a cashier works on a day of the week that they have always been scheduled to work over the past four calendar weeks, they are on average 1.63% more productive than when they work on a day of week that they did not work in the past four calendar weeks. Although these results cover mostly what our initial regression models will test, they do not dive deep into the interaction effect between HOTD and DOTW variables. They found that the coefficient estimate of the interaction is negative and significant, meaning the benefit when one schedule consistency type increases is greatest when the other one is lacking (Lu et al., 2022). However, these interaction results are only one paragraph with no formal hypothesis. For operationalization details including variable construction, rolling window specification, and control variables, see Section 3.1.

Shang et al. (2026) expanded on Lu et al. (2022) by applying the same HOTD and DOTW framework to warehouse workers performing picking and loading tasks, with the added feature of including night shift hours. Since our retail dataset covers only daytime operations, we focus on their findings from regular daytime hours. Shang et al. (2026) replicated the core finding that both HOTD and DOTW consistency improve productivity, but found substantially larger effect sizes than Lu et al. (2022), with DOTW producing gains four to eight times larger than HOTD. The authors attributed this difference in magnitude to a piece-rate pay structure that amplifies workers' behavioral response to consistency, since piece-rate workers have a direct financial incentive to translate any productivity-enhancing condition into greater output. Because our retail dataset operates under hourly pay, we would expect our effect sizes to fall closer to Lu et al.'s original estimates than to Shang et al.'s warehouse findings.

Shang et al. (2026) results are relevant to our study even though the sleep hour boundary falls out of our study since it serves as an alert that we need to expect hour of day effects concentrated in certain hours. Another finding relates to the effect size asymmetry and day of week dominance. Shang et al. (2026) also found that the interaction and its asymmetry, which is most relevant to our paper, replicates Lu et al. (2022)'s negative HOTD x DOTW interaction. Crucially, the interaction is asymmetric. Specifically, they found that DOTW moderates the value of HOTD, but HOTD does not symmetrically moderate the value of DOTW. They noticed that when DOTW is already high, increasing HOTD during regular hours yields no significant benefit. This particular question will be addressed in this paper later on when we test whether HOTD adds meaningful benefit in a retail daytime setting. Additionally, Shang et al. (2026) do not make the interaction their primary research question, which this paper addresses directly.

2.3 Mechanism: Why Consistency Improves Performance

The idea that consistent scheduling improves worker performance is not just intuitive. There is also real empirical evidence for it. Silva and Costa (2023) reviewed 129 studies on the consequences of shift and night work across healthcare, organizational, and family settings. Their sample consisted of a majority of

healthcare workers, and a substantial portion of the findings involved night shifts specifically, which is outside of the context of our retail store. Nonetheless, the core mechanisms they identified, namely circadian disruption, sleep loss, and cognitive impairment, are not unique to hospitals or overnight work. They show up wherever schedules are inconsistent, and retail is no exception.

The most important mechanism Silva and Costa identified is sleep disruption. When workers cannot establish a stable sleep routine because their schedule keeps shifting, they accumulate fatigue over time. That fatigue compounds. Sixteen studies in their review linked schedule inconsistency to anxiety, depression, and burnout, and the authors described a feedback loop where worse sleep leads to worse performance, which creates more stress, which further disrupts sleep. Arguably, a cashier working 7 a.m. shifts three days one week and closing shifts the next is dealing with the same basic biological problem as a rotating night-shift nurse. The scale is different but the mechanism is the same. The body cannot settle into a rhythm, so it never fully optimizes.

Van Dongen and Dinges (2005) help explain the underlying biology. Their two-process model describes how human alertness is governed by two independent systems working simultaneously. The homeostatic process is straightforward: the longer you are awake, the more you need sleep, like a battery running down. The circadian process is different. It runs on a roughly 24-hour internal clock that is largely independent of how tired you are, dictating when your body expects to be alert and when it expects to rest. Performance peaks and valleys track closely with core body temperature, which bottoms out around 4 to 5 a.m. for most people and peaks in the late afternoon. When someone works a consistent schedule, their circadian system aligns to that schedule and their body is genuinely better prepared for cognitive demands during those hours. When the schedule keeps changing, that alignment never happens.

Not everyone's clock runs on the same timing, though. Chronotype, meaning the genetically influenced tendency toward earlier or later sleep-wake patterns, means that a consistent morning schedule will suit some workers much better than others. This is part of why HOTD consistency has heterogeneous effects

and why it is harder to implement as a universal policy lever. DOTW consistency works differently. The biological case for weekly rhythm alignment is weaker than the daily one, but consistent days off and consistent workdays let workers build stable routines around rest and recovery in a way that irregular weekly patterns simply do not allow. The mechanism here is less about the circadian clock and more about behavioral predictability, specifically knowing your week well enough in advance to sleep, eat, and recover consistently.

Taken together, the two dimensions of consistency operate through related but distinct pathways. HOTD is mostly about aligning the body's internal clock to a specific daily window, while DOTW is mostly about anchoring a stable weekly routine. Both ultimately serve the same underlying need: reducing the unpredictability that prevents the body from preparing for work. Because of this, there is good reason to think they overlap enough to produce diminishing returns when both are high at the same time. Whether that is actually what happens is the empirical question this paper sets out to answer.

3. Analysis and Results

3.1 Data and Variables

This study uses a transaction-level dataset of cashier scanning records from a regional grocery retailer operating two store locations. Both locations operated exclusively during daytime hours, which is relevant to interpreting HOTD effects given Shang et al. (2026) finding that HOTD consistency is only beneficial outside of sleep hours. The dataset spans approximately two years, from January 2016 through November 2017, and includes detailed information on transaction timestamps, items scanned, transaction duration, payment methods, cashier identifiers, and total transaction values.

Prior to analysis, several cleaning steps were conducted to ensure data accuracy. While the dataset contained no missing rows, several variables required data type correction and validation. The total transaction value variable was originally stored as a string and was cleaned to remove inconsistent

formatting before being converted to numeric form. Observations with remaining formatting issues, often associated with coupon usage, were removed to preserve data integrity.

Transaction timing variables were carefully validated by reconstructing transaction duration from start and end timestamps and comparing it against recorded values. Observations with implausible durations, those shorter than 1.5 seconds or longer than 900 seconds, were excluded as they likely reflect system errors or interruptions rather than true checkout activity. Duplicate transactions were identified using invoice numbers and, to avoid double counting, only the first occurrence was retained. Extreme values in key variables including transaction time, items per minute, and transactions per hour were winsorized at the 1st and 99th percentiles. Winsorization reduces the influence of extreme observations that may disproportionately affect regression estimates.

Several derived variables were then constructed to support the analysis. A datetime variable enabled time-based grouping, which was used to generate hour-of-the-day and day-of-the-week indicators.

Demand conditions were captured using transactions per hour based on hourly aggregation. A store-terminal identifier was created to distinguish checkout locations across the two stores. A clopening indicator was developed to capture whether a cashier worked a closing shift followed by an opening shift the next day.

The two key independent variables follow the construction established by Lu et al. (2022). Their consistency variables are calculated in a 28-day rolling window. Schedule consistency for hour-of-the-day means how many days in the last 28 days has that employee worked at that particular hour. For day-of-the-week it means how many days in the last 28 days has this employee worked that same day of week. Same as in Lu et al. (2022), the measure for our consistency variables ranges from zero to one, where zero indicates no consistency and one indicates the worker always worked that same slot or day across the prior four weeks. The 28-day window was chosen because it approximates one month of

scheduling history, the period over which workers plausibly develop stable expectations about their routine.

The dependent variable is the natural log of items scanned per minute for each transaction. The log transformation is appropriate because the raw distribution of items per minute is right-skewed, and the transformation allows coefficients to be interpreted as approximate percentage changes in productivity.

After all cleaning and variable construction steps, the final analytical sample consists of 996,569 transactions processed by 126 cashiers.

3.2 Methods

The study this paper builds upon used transaction-level data from two grocery store locations (Lu et al., 2022). Their data includes 1,238,418 transactions by 126 cashiers from 2016 to 2017. Their consistency variables are calculated in a 28-day rolling window. Schedule consistency for hour-of-the-day means how many days in the last 28 days has that employee worked at that particular hour. For day-of-the-week it means how many days in the last 28 days has this employee worked that same day of week. We will replicate their operationalization as we will validate their results and from there build further the interaction effect.

This study tests whether hour-of-the-day (HOTD) and day-of-the-week (DOTW) schedule consistency interact to affect cashier productivity. While Lu et al. (2022) found that each consistency type independently improves performance, they reported a negative interaction between the two as a secondary finding without developing managerial implications. We make this interaction our primary research question: do these two consistency types work together, or does one substitute for the other?

To answer this question, we use fixed-effects regression analysis following the interaction testing framework established by Aiken and West (1991). We estimate a series of models that build from examining each consistency type separately to testing their combined effect, culminating in a model that

includes a multiplicative interaction term. This approach allows us to determine whether the productivity benefit of HOTD consistency depends on the level of DOTW consistency, or vice versa.

In all models, we treat DOTW as the primary effect and HOTD as the moderating variable, consistent with Shang et al. (2026) finding that DOTW produces larger and more stable effects across conditions.

Regression Model

Our primary estimating equation takes the form:

$$\log(\text{Items Per Minute}) = \beta_1(\text{HOTD}) + \beta_2(\text{DOTW}) + \beta_3(\text{HOTD} \times \text{DOTW}) + \text{Controls} + \text{error}$$

The parameter of interest is β_3 , the interaction coefficient, which tells us how the two consistency types relate. If β_3 is negative, the two types function as substitutes, meaning there is no synergic effect. Once a worker has high consistency in one dimension, adding consistency in the other provides less additional benefit. If β_3 is positive, the two types are complements, meaning they reinforce each other and both are needed to maximize productivity. If β_3 is not statistically significant, it suggests that the two types work under different underlying mechanisms and their effects might be additive. Based on Lu et al.'s exploratory finding and our theoretical framework, which holds that both consistency types stabilize worker rhythms through partially overlapping mechanisms, we hypothesize β_3 less than 0, indicating a substitute relationship.

We estimate four models sequentially. Model 1 isolates the control variables, not including consistency measures, allowing us to gain a baseline understanding of how fixed effects influence our model. Model 2 includes HOTD consistency only, replicating Lu et al.'s main effect for hour-of-the-day consistency. Model 3 includes DOTW consistency only, replicating Lu et al.'s main effect for day-of-the-week consistency. Model 4 adds the interaction term to the joint specification, which is our primary model for testing the substitute hypothesis. This sequential approach lets us observe how the interaction changes our understanding of the main effects and whether it adds meaningful explanatory power.

Control Strategy

All models control for factors that might confound the relationship between schedule consistency and productivity. We include demand conditions measured as transactions per hour to capture shift intensity and workload. We control for transaction characteristics that mechanically affect scanning speed, including basket size and payment method indicators. We include a clopening dummy that equals one when a worker closes one night and opens the next morning, capturing potential fatigue effects. Fixed effects are included for cashier identity, hour of day, day of week, and store-terminal combination. Cashier fixed effects absorb individual skill differences, hour-of-the-day effects account for rush versus slow periods, day-of-the-week effects account for weekend versus weekday traffic patterns, and store-terminal effects account for equipment differences across checkout lanes. Standard errors are clustered at the cashier level because each worker appears in multiple transactions, creating correlated errors within individuals.

3.3 - Analysis and Results

Table 1. Impact of Schedule Consistency on Cashier Productivity

Coefficient estimate / percentage of cashier productivity				
Dependent Variable	No X	HOTD Model	DOTW Model	Interaction Effect
HOTD	-	0.0282	-	0.0644
DOTW	-	-	0.0283	0.0570
HOTD x DOTW	-	-	-	-0.0622
Cloping	0.0026	0.0049	0.0050	0.043
C(day_of_week)[T.2]	-	-	0.0036	0.0047

(Tuesday)				
C(hour)[T.14] (2 pm)	-	-0.1527	-	-0.1508
C(nitem)[T.83] (Item 83)	2.4727	2.4718	2.4730	2.4757
C(employeeid)[T.2284]	-0.4029	-0.3680	-0.3781	-0.3701
Various additional controls	-	-	-	-
R ²	0.325	0.328	0.326	0.328

Table 1 presents results across all five model specifications. Because the dependent variable is in log form, coefficients are interpreted as approximate percentage changes in productivity.

Interpretation

For the interaction model, we interpret the coefficient on HOTD x DOTW to assess the direction and magnitude of the relationship between the two consistency dimensions. A negative coefficient confirms a substitute relationship. The practical implication is that managers facing scheduling constraints should prioritize improving whichever consistency dimension is currently weaker, since the marginal return from adding consistency is highest when the other dimension is low.

Main Effects

The baseline model, which excludes the consistency variables, provides a useful benchmark for understanding how much of cashier productivity can be explained by time controls and transaction-level factors alone. In this specification, the model achieves an R² of 0.325, indicating that these controls capture a substantial portion of the variation in productivity. The coefficient on clopening is small and

statistically insignificant (0.0026), suggesting that working a closing shift followed by an opening shift has no meaningful standalone effect on performance. Overall, this baseline establishes that while much of the variation is explained by observable factors, there remains room to assess whether scheduling structure contributes additional explanatory power. The relatively high R^2 in this baseline also underscores that any gains from adding consistency measures should be interpreted as incremental improvements on an already well-specified model.

Building on this baseline, the next model introduces hour-of-the-day (HOTD) consistency alongside calendar-date fixed effects to more rigorously evaluate the role of schedule regularity. By incorporating these date controls, we are able to isolate the effect of consistency from day-specific shocks such as demand fluctuations or store-level events, providing a cleaner estimate of how repeated exposure to the same working hours influences cashier productivity. In this case, changes in R^2 relative to the baseline (0.325) help quantify the incremental explanatory power gained from adding both HOTD consistency and calendar-date fixed effects.

Model 3 shows that DOTW consistency alone is associated with a 2.8% increase in productivity (coefficient = 0.0283, $p < 0.01$), nearly identical in magnitude to the HOTD estimate from Model 2. The R^2 of 0.326 is comparable to the HOTD models, confirming that both dimensions capture meaningful variation in worker performance. Taken together, Models 1 and 3 establish that each form of consistency independently contributes to higher productivity, consistent with prior findings in the literature

Interaction Model

The interaction specification serves as the primary model for understanding how the two dimensions of consistency relate to one another. Both HOTD (0.0644) and DOTW (0.0570) remain positive and statistically significant, confirming that each dimension is independently associated with higher cashier productivity. However, the interaction term between HOTD and DOTW is negative and statistically

significant (-0.0622), supporting the hypothesis that the two dimensions act as substitutes rather than complements.

Treating DOTW as the moderating variable, this result indicates that the productivity benefit of HOTD consistency diminishes as DOTW consistency increases. When a cashier already has a stable day-of-the-week schedule, the marginal gain from also stabilizing their hour-of-the-day schedule is reduced. Conversely, when day-of-the-week consistency is low, HOTD consistency plays a more substantial role in improving productivity. The combined effect of both dimensions remains positive, but is smaller than the sum of their individual contributions, indicating diminishing returns when both are simultaneously high. This aligns with the theoretical expectation that both forms of consistency operate through overlapping behavioral and biological mechanisms.

The larger main effect coefficients relative to the single-variable models reflect standard interaction model behavior, where coefficients capture effects conditional on the other variable. The R^2 of 0.328 represents only a marginal improvement over prior specifications, suggesting that while the interaction term adds meaningful interpretive insight into the structure of the relationship, it does not substantially increase overall explanatory power. The clopening variable remains small and statistically insignificant, indicating no meaningful relationship between opening-after-closing schedules and productivity once other factors are controlled for. In addition to the primary variables of interest, the inclusion of time fixed effects and transaction-level controls in this specification further strengthens the interpretation of the results.

Hour-of-the-day and day-of-the-week fixed effects capture systematic variation in productivity driven by demand fluctuations and operational conditions, while transaction size (as proxied by item count) exhibits a large positive association with productivity, consistent with economies of scale in checkout processes. Employee fixed effects account for persistent differences in cashier ability, ensuring that the estimated effects of schedule consistency are not driven by inherently more productive workers receiving more stable schedules. Together, these controls isolate the impact of schedule consistency from temporal, operational, and individual-level confounders, reinforcing the robustness of the interaction results.

Comparison to Prior Literature

Our main effect estimates are somewhat larger than those reported by Lu et al. (2022), who found HOTD associated with a 0.95% gain and DOTW with a 1.63% gain. Our estimates of approximately 2.8% for both dimensions in the single-variable models are directionally consistent but larger in magnitude, which may reflect differences in the observation window, store context, or workforce composition. The interaction coefficient of -0.0622 is also larger in absolute value than Lu et al.'s reported interaction of -0.01856, pointing to a stronger substitute relationship in our data. Both findings are directionally aligned with prior work and with the theoretical prediction of diminishing returns when both consistency dimensions are simultaneously high.

3.4 Robustness Checks

To ensure our models' reputability, we ran a robustness check by re-estimating all models using a shorter 14-week rolling window in place of the baseline 28-week window used to construct the consistency measures. This adjustment tests whether the observed relationships are sensitive to the time horizon over which consistency is defined. The results remain highly consistent across specifications, with nearly identical model fit (R^2 ranging from 0.325 to 0.328) and stable coefficient magnitudes for HOTD, DOTW, and their interaction. In particular, both consistency measures retain positive effects, while the interaction term remains negative, reinforcing the substitution effect observed in the primary specification.

Taken together, these findings provide strong evidence that the relationship between schedule consistency and productivity is stable across model specifications and robust to alternative constructions of the key independent variables. The consistency of both coefficient estimates and model fit suggests that the results are not driven by specific modeling choices, but rather reflect a persistent and structurally meaningful relationship in the data. Additional robustness checks could include testing alternative outlier thresholds, using different functional forms for productivity, and estimating the model across subsamples such as peak versus non-peak hours or high- versus low-volume cashiers. We focused on varying the

rolling window used to construct consistency measures since this directly tests the sensitivity of our key independent variables in regards to the time horizon over which scheduling patterns are defined, rather than altering secondary modeling assumptions.

4 - Discussion

Our findings demonstrate that schedule consistency is a meaningful predictor of cashier productivity. Both hour-of-the-day consistency and day-of-the-week consistency variables exhibit positive associations with items scanned per minute after controls are applied. These findings suggest that consistent work scheduling enhances worker performance by limiting disruptors and allowing them to adapt more effectively to their work schedules. Even though the coefficients are modest in size, they remain important because they persist across specifications that account for cashier heterogeneity, demand conditions, transaction complexity, and terminal effects. Overall, the results support the thesis that more stable scheduling arrangements are not merely beneficial for workers' routines, but are also operationally advantageous for retail performance. We also checked whether these results depend on how we defined consistency, and when we re-estimated the models with alternative rolling windows, the signs and overall fit stayed almost the same. Our substitution findings appear robust.

The additive model that includes both HOTD and DOTW consistency without interaction provides an important insight. In this model both consistency dimensions remain positive and statistically significant, showing that when considered together, each independently contributes to higher cashier productivity. This can be interpreted as evidence that both hour-level and day-level regularity enhance worker performance through distinct mechanisms. Each stabilizes one form of routine, either daily biological rhythms or weekly behavioral routines. It is also worth noting that this model's explanatory power does not substantially exceed that of the single variable models, which leads us to believe that their effects overlap meaningfully. This overlap suggests that while both consistency types matter for productivity, they may influence aspects of worker adaptation through partially shared channels rather than entirely

independent ones. This model helps us understand why the interaction later shows diminishing efficiency returns when both are high.

Our interaction results suggest that hour-of-the-day consistency and day-of-the-week consistency do not simply add to one another in a linear way. Instead, the benefit of one form of consistency depends on the level of the other. The negative interaction coefficient indicates a substitutive relationship. This means that once a cashier is already highly consistent in one dimension, the marginal gain from increasing the other dimension becomes smaller. This is consistent with the idea that schedule stability works through overlapping mechanisms of consistency rather than two independent channels. A schedule that is already stable by day may reduce the additional value of further stabilizing the hour, and vice versa.

Our interaction models also show that both consistency measures remain positive and statistically significant, meaning that both matter even when considered jointly. However, there is a diminishing return implied in our interaction that suggests managers should not treat HOTD and DOTW as equally contributing levers. Instead, managers should think of them as partially overlapping inputs into productivity.

The interaction findings strengthen the theory that schedule consistency improves productivity through both biological and behavioral stabilization, but these mechanisms are not perfectly independent. HOTD seems to be more closely tied to circadian alignment while DOTW is more related to routine formation and predictability. The negative interaction suggests these overlap enough to produce diminishing marginal returns when combined. This refines our understanding of consistency as a structured set of dimensions with interdependent effects.

The results also extend the literature on rhythm synchronization by suggesting that workers may use one type of regularity to compensate for partial weakness in the other. This is theoretically important as it implies a substitute relationship rather than a complementary one. This challenges the assumption that all

forms of predictability are equally additive, and it supports the view of how work schedules shape human performance.

For managers, the main takeaway is that consistency should be designed strategically, not uniformly. If the goal is to improve checkout productivity, the evidence suggests that firms should first stabilize whichever consistency dimension is more feasible to control in their setting. For many retailers, day-of-the-week stability may be easier to standardize across workers, while hour-of-the-day consistency may require more careful alignment with demand peaks and labor availability. Managers should also recognize that the gains from adding more consistency may taper off once one dimension is already strong.

This has practical implications for schedule posting. Rather than spreading flexibility evenly across all employees, managers can concentrate on consistency where it will produce the largest marginal gain. The results also imply that scheduling policies should not focus only on fairness or predictability, but on the operational design of stable patterns that fit the store's labor needs.

To quantify the practical impact of schedule consistency, we simulate improvements in hour-of-day (HOTD) and day-of-week (DOTW) consistency using the interaction model and average employee-level baseline values (HOTD = 0.440, DOTW = 0.603). Both dimensions are positively associated with productivity, though a negative interaction effect indicates diminishing returns when they are improved simultaneously.

Individually, a 10 percentage point increase in HOTD consistency results in a 0.27% productivity gain, translating to approximately 9 hours of labor savings per employee annually ($\approx 1,129$ hours at the store level). A comparable increase in DOTW consistency yields a slightly larger 0.30% improvement, or roughly 10 hours per employee annually ($\approx 1,243$ hours total), showing how DOTW is the stronger independent lever.

When both dimensions are improved together, the gains increase but are not fully additive. A simultaneous 10% improvement in both HOTD and DOTW produces a 0.50% productivity gain, equivalent to approximately 16.8 hours per employee ($\approx 2,100$ hours store-wide). Increasing both to 15% further raises productivity by 0.71%, generating approximately 23.6 hours per employee annually ($\approx 3,000$ hours total).

Targeted, asymmetric improvements also yield strong results. Increasing HOTD by 10% and DOTW by 15% results in a 0.62% productivity gain (≈ 20.7 hours per employee; $\approx 2,600$ hours total), while the reverse allocation (HOTD +15%, DOTW +10%) produces a slightly smaller 0.60% gain (≈ 20.2 hours per employee; $\approx 2,546$ hours total). This suggests that prioritizing DOTW consistency may offer marginally higher returns at current baseline levels.

Overall, while diminishing returns are present, coordinated improvements across both dimensions—or strategically weighted combinations—produce the largest operational gains. These findings highlight schedule consistency as a low-cost, high-impact lever for improving workforce productivity and expanding effective labor capacity without additional staffing.

It is important to consider key limitations in our study. We utilize observational retail transaction data, which suggest results show association rather than definitive causation. The sample is also drawn from a single retail grocery chain, which may limit generalizability to other retail formats, such as unions, wage systems, or labor markets. Additionally, the interaction is estimated within a specific operational context, so the magnitude may differ in settings with stronger demand volatility or different staffing structures.

Future research should test whether the same interaction pattern holds in other sectors, with particular interest in those with more irregular schedules or night work. Additionally, replicating these findings using alternative window specifications, such as a 14-day rolling window, would help establish whether the substitute relationship is robust to the construction of the consistency variables. Finally, researchers could build on these findings to develop formal optimization frameworks for scheduling consistency

allocation under real staffing constraints, turning the interaction results into a more direct managerial decision tool.

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